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Course: Calculus Beta

Assignment: MyLab Math - Uge 4

Graph the points $z = x + iy$ that satisfy the given condition.

$$|z + 8| \leq |z - 8|$$

Rewrite the given condition so that both sides are in the same form.

$$\begin{aligned} |z + 8| &\leq |z - 8| \\ |z - (-8)| &\leq |z - 8| \quad \text{Rewrite the left side.} \end{aligned}$$

Each side of the given condition is now in the form $|z_1 - z_2|$. This expression represents the distance between the two points z_1 and z_2 in the complex plane.

This means the points that satisfy the condition $|z - (-8)| \leq |z - 8|$ are the points whose distance from -8 is less than or equal to their distance from 8 .

Identify these points in two phases. First find the points whose distance from -8 is equal to their distance from 8 .

In the complex plane, -8 and 8 correspond to the points $(-8, 0)$ and $(8, 0)$, respectively.

The goal for this phase is therefore to find the points that are equidistant from these two points, which are shown in the graph. These points are precisely those on the y -axis.

This means all the points on the y -axis satisfy the given condition. This is shown in the graph. Note that the line is solid because the points on it satisfy the condition.

Now find the points whose distance from -8 is less than their distance from 8 .

These points are those that are closer to $(-8, 0)$ than $(8, 0)$, meaning they are to the left of the y -axis.

This means all the points to the left of the y -axis also satisfy the given condition. This is shown in the graph.

Thus, the set of points that satisfy the given condition consists of the y -axis and all points to the left of it. This is shown in the graph.

